

TRACE — DIGITAL IMAGE FORENSICS

Every Pixel Leaves A Trace.

Problem: The increasing use of image-editing and generative AI tools makes it difficult to determine whether digital images have been manipulated.

Our Solution: A web-based platform that performs preliminary forensic analysis of images and highlights potential signs of manipulation.

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Problem & Objectives

Problem Statement

Manipulated and AI-generated images can be used for misinformation, impersonation, fraud and digital abuse. Manual inspection is often insufficient to identify subtle alterations.

Objectives:

- Analyse images using multiple forensic techniques.
- Identify suspicious regions and characteristics.
- Provide an understandable preliminary assessment.
- Present forensic evidence visually.
- Help users make more informed judgements about digital images.

How TRACE Works

01

IMAGE UPLOAD

02

PRE-PROCESSING

03

FORENSIC ANALYSIS

04

ANOMALY DETECTION

05

COMBINED ASSESSMENT

06

VISUAL RESULTS

TRACE does not depend on a single detection method. It combines multiple image characteristics to produce a preliminary manipulation likelihood.



Note: The result is an assessment, not legal or definitive proof of manipulation.

Techniques Used

Metadata Analysis

Examines available information about the image and its creation/editing history.

Error-Level Analysis (ELA)

Highlights areas that may have different compression characteristics.

Noise Analysis

Identifies inconsistencies in image noise patterns.

Frequency Analysis

Examines frequency-domain patterns that may reveal unusual image characteristics.

Edge & Texture Analysis

Checks structural and texture inconsistencies.

Anomaly Detection

Combines forensic indicators to identify potentially suspicious regions.

Technology & Architecture

Technology Stack

- Frontend: React, Next.js, TypeScript, Tailwind CSS
- Backend: Python, FastAPI
- Image Processing: OpenCV, Pillow, NumPy
- Visualisation: WebGL, interactive forensic visualisations
- API: REST API integration
- Development: Antigravity IDE, Cursor, VS Code, Git

Architecture

01

USER

02

TRACE WEB UI

03

FASTAPI BACKEND

04

ELA + METADATA + NOISE

05

FREQUENCY / ANOMALY ANALYSIS

06

RESULTS

What TRACE Provides

- Overall preliminary authenticity assessment
- Suspicious-region identification
- ELA and noise visualisation
- Frequency-domain analysis
- Metadata information
- Explainable forensic indicators

Innovation / Extra Effort

- Multiple forensic techniques in one platform
- Interactive visual forensic analysis
- Accessible web-based workflow
- ADRIA assistant for explaining results
- Multilingual and voice-based assistance
- Extensible architecture for external detection APIs

Conclusion & Future Scope

Conclusion

TRACE provides an accessible approach to preliminary digital image forensics by combining multiple forensic techniques and presenting their findings in an understandable form.

Future Scope

- Advanced AI/deepfake detection models
- Video and audio forensics
- Improved machine-learning-based anomaly detection
- Larger forensic datasets
- Mobile support
- Expanded multilingual and voice capabilities